Geophysical Journal (1988) **94,** 157–166

A palaeomagnetic study of a Caledonian ophiolite

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Accepted 1988 February 3. Received 1988 January 25; in original form 1987 February 26

SUMMARY

A total of 89 sites have been sampled for palaeomagnetic studies from the Caledonian ophiolite complex of northeast Shetland (60.7°N 359.2°E) and a post-ophiolite emplacement suite of lamprophyre dykes. Of these only 35 sites from the ultramafite facies of the complex yielded statistically significant results. These sites define three distinct populations of magnetization directions which correspond to south palaeomagnetic poles at 47.2°S 244.2°E (*A*), 18.0°S 323.4°E (B), and 35.0°S 351.1°E (*C*). Of these only the *C* remanence can be assigned an unambiguous age of mid-Carboniferous; the ages of the other two poles remain ambiguous due to the uncertainty in the APW path and the complexity of the ophiolite history. Nevertheless palaeomagnetism is shown to be successful in detecting low temperature, low pressure, alteration events that have affected the complex and which hitherto have gone undetected.

Key words: palaeomagnetism, palaeozoic, Shetland ophiolite complex

1 INTRODUCTION

This study forms part of a wider geophysical study of the nature and tectonic history of the NE Shetland ophiolite complex (Taylor 1985). The intention of the study reported here was to use palaeomagnetism to detect, and chronologically resolve, the tectonothermal events which have affected the complex during it's emplacement and subsequent history. While the results presented in the body of the paper will show that such events can be detected by palaeomagnetism, the main obstacle to dating them remains the fact that the APW path for this part of the Scottish Caledonides is still not sufficiently resolved to allow a complete dating of the individual palaeomagnetic poles.

2 GEOLOGICAL OUTLINE AND AGE OF THE COMPLEX

The complex is exposed in the east of the islands of Unst and Fetlar which are part of the Shetland archipelago which lies to the northeast of Scotland and west of Norway. The complex consists essentially of two ophiolitic thrust nappes separated from each other and the underlying basement rocks by mélanges of both tectonic and sedimentary origin (Fig. 1). The basement to the area comprises heterogeneous schists and gneisses which have been equated with the Lower Dalradian Succession of Scotland (Flinn *et al.* 1972). This basement is overlain by thrust sheets composed of metasediments (the Saxa Vord block of Fig. 1) which have been interpreted (Flinn 1958) as being emplaced prior to the ophiolite complex thrust sheets. The lower ophiolitic thrust nappe contains a classic ophiolite sequence comprising mantle harzburgite overlain in turn by a transitional layered harzburgite–dunite unit, massive dunite, clinopyroxenite (often a wehrlite) and finally gabbro. Many of these units are banded or layered both at the main lithological boundaries and internally. In addition dolerite dykes, plagigranites and metavolcanics have all been recorded from the lower thrust nappe to provide a complete representation of all the members of a normal ophiolite sequence. The upper ophiolitic thrust nappe consists predominantly of harzburgite, with occasional lenses of dunite, which is commonly found to be banded or layered. No simple structural model has yet emerged for the complex as a whole. On Fetlar it would appear that both the thrust planes carrying the upper and lower thrust nappes have been synformally folded about a N–S axis (Flinn 1958) while on Unst the structure is much less clear, the upper nappe seeming to rest on the lower nappe as an isolated klippe. The thrust boundary separating the lower thrust nappe from the underlying mélange and basement on Unst is observed to dip very steeply eastward so that the outcrop is near linear in the southern part of the island. In contrast it becomes curvilinear in the northern sector of it's outcrop and parallels topography, due presumably to a shallowing of the angle of contact with the underlying rocks.

The oldest radiometric age yet recorded from the complex is 496 ± 21 Ma from a garnet bearing amphibolite from the metamorphic sole of the upper nappe on Fetler, the analysis was made on a whole rock sample using the ^{40}Ar–^{39}Ar method (Taylor 1985). This age suggests that obduction took place in late Cambrian or early Ordovician times which is consistent with the general view that the Iapetus ocean may have started to close by these times. There is however

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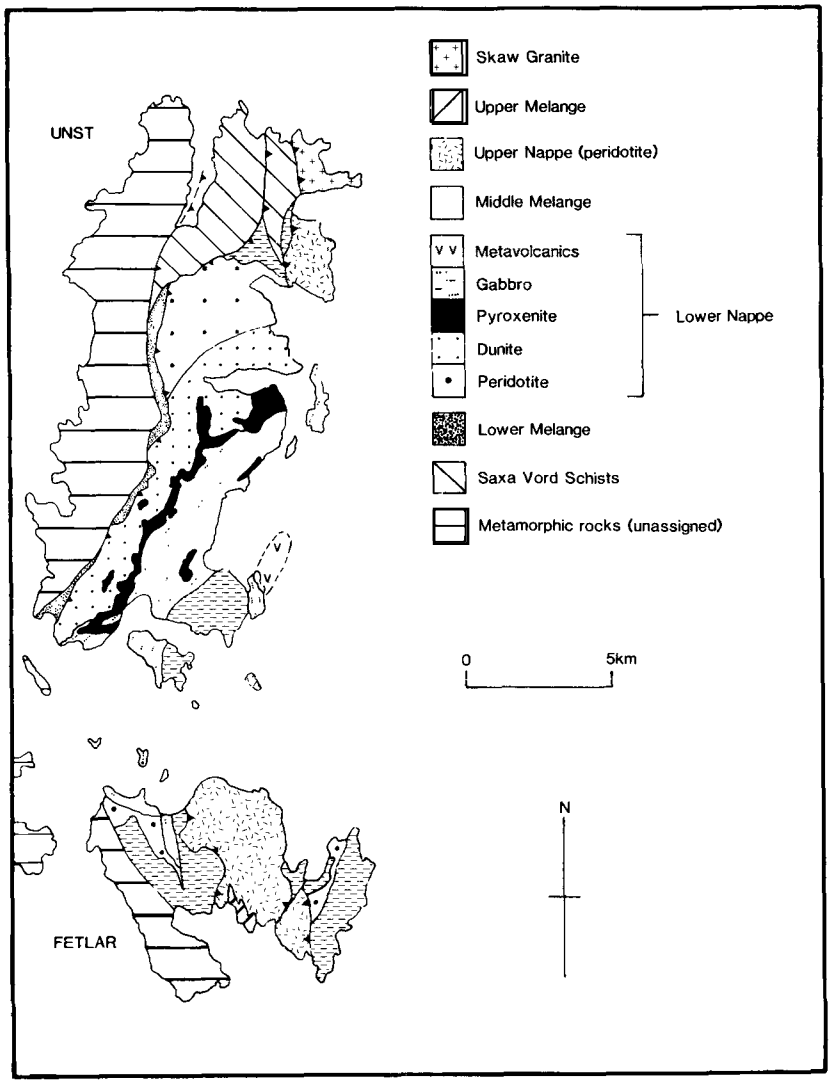


Figure 1. Geological sketch map of the NE Shetland ophiolite complex.

no conclusive evidence, at this time, of the original tectonic setting of the ophiolite. Both K–Ar (Miller & Flinn 1966) and ^{40}Ar–^{39}Ar (Taylor 1985) dating of samples from the Dalradian basement, the mélanges and the basic rocks of the complex have yielded ages in the range 420–430 Ma. These ages may be linked to a tectonothermal resetting of the isotopic systems by the stacking of the nappes which is recorded as a widespread greenschist grade burial metamorphism which affects the sediments in the mélanges between the thrust sheets, the basement on which they rest, and the nappes themselves. This metamorphism takes the form of a tectonising event in the sediments and shear zones within the thrust sheets, while static recrystallization has also lead to the formation of chloritoid in basement rocks of suitable composition and the production of antigorite bearing serpentinites at the base of the ultramafite units bounded by thrusts (Flinn 1985). The harzburgite and dunite rocks have suffered extensive serpentinisation, which is estimated to be 70–100 per cent from the measured densities of rock samples prior to this recrystallization. The terminal event in the emplacement history of the complex would appear to be a late stage tectonisation leading to widespread cataclasis and shearing; this has affected some of the dykes in a lamprophyre dyke suite emplaced during and after this cataclasis. Two dykes from this latter suite have yielded ^{40}Ar–^{39}Ar ages of 407 ± 7 Ma and 418 ± 10 Ma, the latter determination was from a dyke showing schistose margins, and suggests that late stage tectonic events continued at least until this time.

3 EXPERIMENTAL PROCEDURES AND RESULTS

3.1 Methodology

A total of 89 sites were collected from all facies of the complex and the lamprophyre dyke suite which cuts both the basement and the complex itself (Fig. 2 and Table 1). Each site normally comprised at least six individually orientated cylindrical cores of 25 mm diameter collected with a portable drill. The field orientations were determined with a sun and/or magnetic compass in conjunction with topographic sighting. A parastatic magnetometer was used

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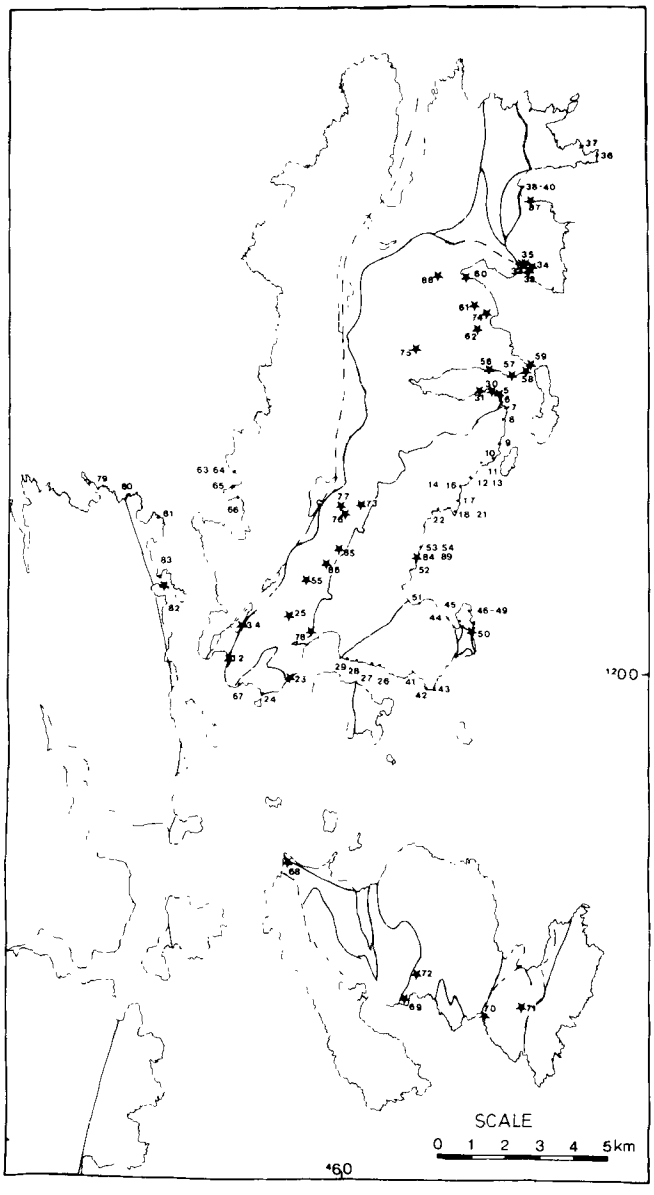


Figure 2. Map showing the palaeomagnetic sampling sites and principle geological boundaries. Those sites yielding significant results are denoted by asterisks and site numbers.

Table 1. Summary of sampling sites.

Lithology	No. of sites	Location
Antigorite	6	Lower thrust nappe of Unst
Harzburgite	5	
Dunite	6	
Pyroxenite	9	
Gabbro	13	
Sheeted dykes	2	
Volcanics & dykes	6	
Harzburgite/dunite	5	Upper nappe of Unst
Ultramafites	5	Upper and lower nappes of Fetlar
Lamprophyre dyke swarm	33	Throughout Unst and NE Yell

to measure the remanent magnetization of the samples and a two-axis tumbling alternating field (AF) demagnetizer or a Schonstedt TSD-1 oven used for the demagnetization studies. AF demagnetization was adopted as the routine method of treatment because it was found to be more efficient in discriminating between individual magnetizations as it yielded better defined linear trajectories on orthogonal plots in comparison to thermally demagnetised sister samples which tended to yield curved demagnetization trajectories and because it more readily removed random low stability components. The applied peak field in AF demagnetization was incremented in steps of 5 or 10 mT up to a maximum possible field of 160 mT while progressive thermal demagnetization was carried out in steps of 50 °C up to 450 °C or 500 °C and 20 °C–30 °C steps above this temperature and beyond the Curie points. In addition a SQUID magnetometer system (Shaw, Share & Rogers 1984) was used to investigate the remanence of weak samples and a variety of equipment was used to perform rock magnetic experiments to determine the properties of the magnetic carriers.

3.2 Results

With the exception of three sites, which show poor within site grouping but are nevertheless significant at the 95 per cent confidence level, the gabbros and sheeted dykes of the complex and the post-ophiolite emplacement lamprophyre dyke swarm proved to be unsuitable for palaeomagnetic analysis on aecount of their weak and/or unstable magnetic moments. All of these rocks have suffered late stage greenschist metamorphism which has produced chlorite as the dominant Fe-bearing phase; this metamorphism may have been responsible for the alteration of any original magnetite present. In contrast, the serpentinised ultramafites tend to have large NRM intensities, often in excess of 1 Am-1, and high bulk susceptibilities reflecting the relatively high modal content (1–5 per cent) of magnetite in these rocks. Magnetite is optically observed to occur in three common forms namely, (i) large skeletal grains, (ii) fine dusts and clusters of small grains in association with serpentine alteration products after olivine and pyroxene, and (iii) as irregular trains of crystals in serpentine veins which cross-cut each other and all earlier fabrics.

Demagnetization studies reveal three distinct characteristic directions, referred to as *A*, *B* and *C* (see Section 4). The bulk of samples were subjected to systematic AF demagnetization treatment, as opposed to thermal demagnetization, as this proved to be more successful in separating components. The vast majority of samples had median destruction fields (that is the peak AF demagnetization field at which their moment is reduced to 50 per cent of the total NRM) in the range 5–30 mT (Fig. 3), although directional stability typically continued up to fields of 50–70 mT and less frequently up to 100–150 mT. These samples thus possess large low coercive components of magnetization, due presumably to a large amount of multidomian grain size magnetite, but nevertheless they carry a residual stable magnetization. Randomly directed low coercivity components were commonly detected but were always removed on treatment to 20–30 mT or less (Fig. 3). Orthogonal plots typically only show one distinct straightline segment after

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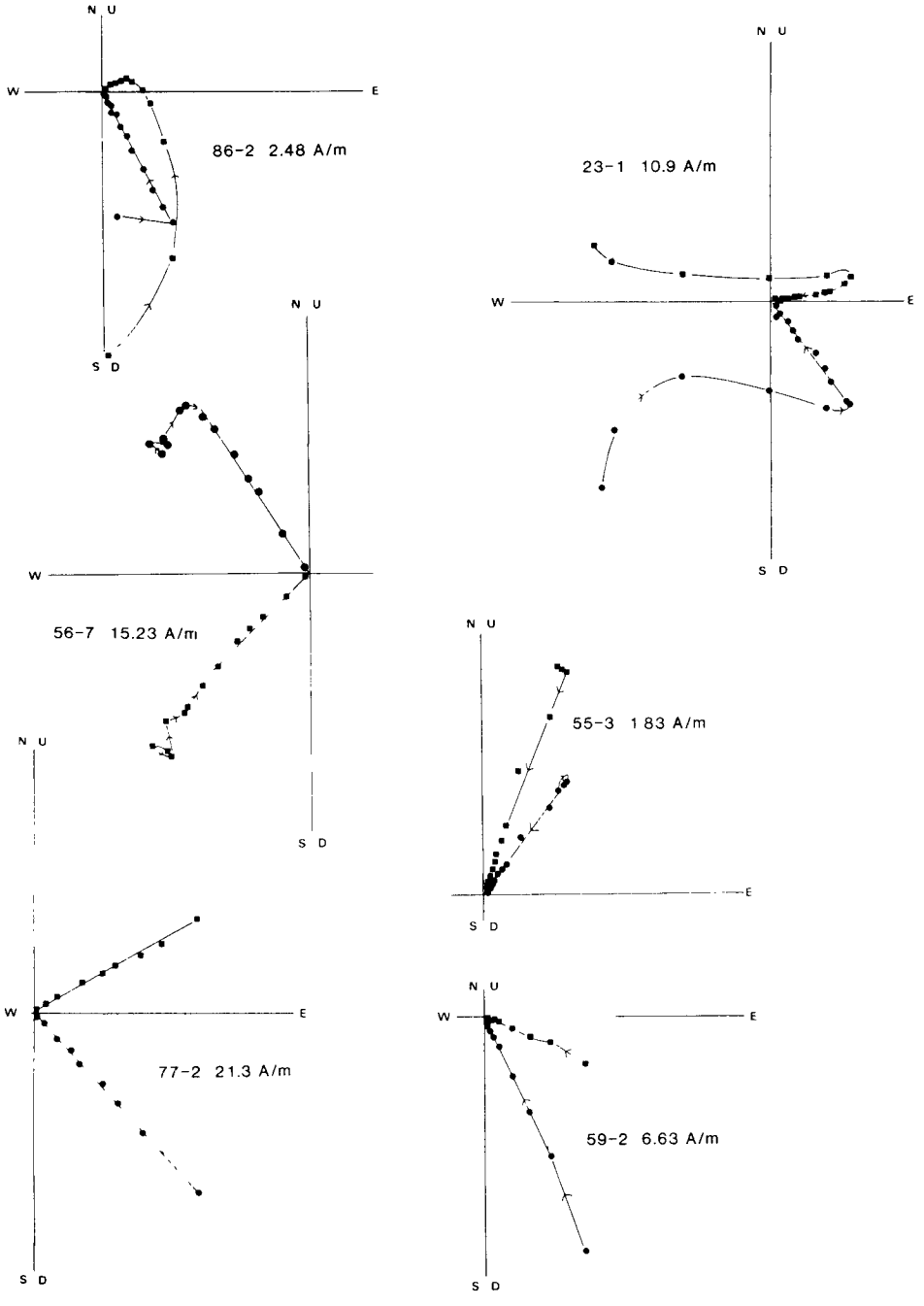


Figure 3. Demagnetization behaviour of typical samples: orthogonal projections of *in situ* NRM directions onto the horizontal (■) and vertical N–S (●) planes. The demagnetizations were carried out in 5 mT steps up to 50 mT and in 10 mT steps above this. Total NRM intensities are given for each sample in Am^{-1}. Sample 56–7 was thermally demagnetized in steps of 50 °C up to 500 °C, then at 530 °C and 560 °C above which it was unstable.

the removal of any overlapping low coercivity components and no systematic differences in the coercivity of the three characteristic directions was identified.

At least one and commonly two or three samples from each site were subjected to thermal demagnetization. The majority of samples showed a continuous decrease in intensity of magnetization in the temperature range 100°–500 °C and a subsequent rapid decrease in intensity (e.g. samples 56–7, 74–5 in Fig. 4). Some samples however demonstrated greater resistance to thermal demagnetization (e.g. 62–6 Fig. 4) or an intermediate form of behaviour with an initial loss of intensity up to 250 °C and then a more gradual loss of intensity up to about 500 °C prior to a rapid decrease in intensity again. Most samples showed similar directional changes to AF treated counterparts but produced more scattered results or incomplete resolution of individual components. Blocking temperature spectra varied widely between sites and was not apparently correlated to either lithology nor the magnetization component carried by the particular sample.

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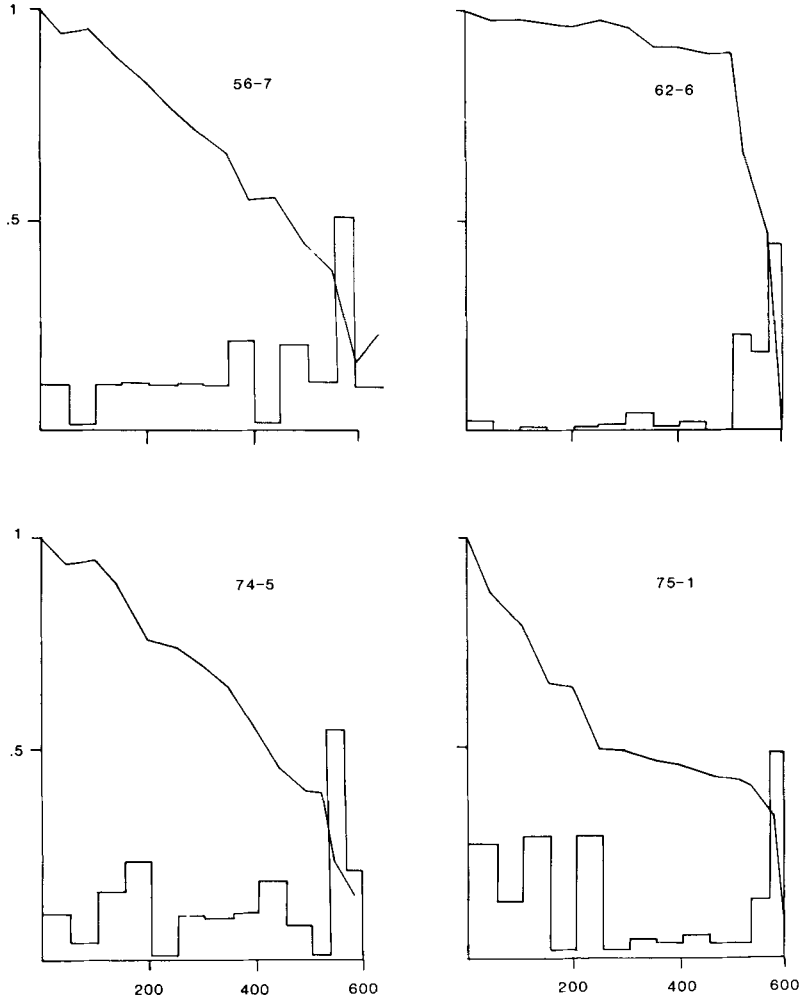


Figure 4. Thermal decay curves for some dunite samples which demonstrate the range of behaviour encountered in thermal demagnetization. The blocking temperature histograms were calculated assuming the simplifying assumption of only one component.

3.3 Rock magnetic experiments

Since no sytematic differences in the nature of the magnetic properties of samples carrying the different characteristic directions were detected during routine demagnetization studies, a number of rock magnetic experiments were performed to discern any subtle differences in the magnetic properties of samples carrying these different magnetizations. These experiments included the determination of Curie points by thermomagnetic investigations, the acquisition of isothermal remanence (IRM), the modified Lowrie–Fuller grain size test (Lowrie & Fuller 1971; Dunlop, Hanes & Buchan 1973; Johnson, Lowrie & Kent 1982), and low temperature susceptibility measurements (Senanayake & McElhinny 1981, 1982; Radhakrishnamurty *et al.* 1982).

Thermomagnetic curves were determined from at least one sample from each site. The typical response for all the ultramafite rocks is that the heating curve is concave upward to temperatures of 300 °C or so, and then becomes convex with a Curie point in the range 560–580 °C (e.g. samples 75–3, 1–1, 4–3 in Fig. 5). Cooling curves were always found to be irreversible with room temperature partial saturation moments normally in the range 0.5–0.7, although rare exceptions were found (e.g. 62–3 in Fig. 5), of the original moment indicating the partial destruction of the magnetic carrier (or conversion to a less magnetic phase) during the heating and cooling cycle (Fig. 5). Only two samples gave any indication of the presence of hematite (3–1 and 88–5 in Fig. 5) which may have formed during heating. This behaviour is consistent with that observed during thermal demagnetization and may suggest a thermochemical origin for the magnetite remanences.

IRM acquisition curves (Fig. 6) all show rapid saturation of moment between 150 and 200 mT and give no indication of the presence of hematite. A total of 22 samples were subjected to the modified Lowrie–Fuller test in which the AF stability of NRM, IRM and an artificially induced anhysteric magnetization (ARM) are compared and all samples yielded results indicating single domain (sd) or pseudosingle (psd) grain size magnetite to be the dominant magnetic grain size contributing to the magnetic properties. Such an indication of the dominance of sd or psd grain size was not expected, both because the NRM coercivities typically had low MDFs, suggestive of md grain size magnetite and because of the abundance of optically large

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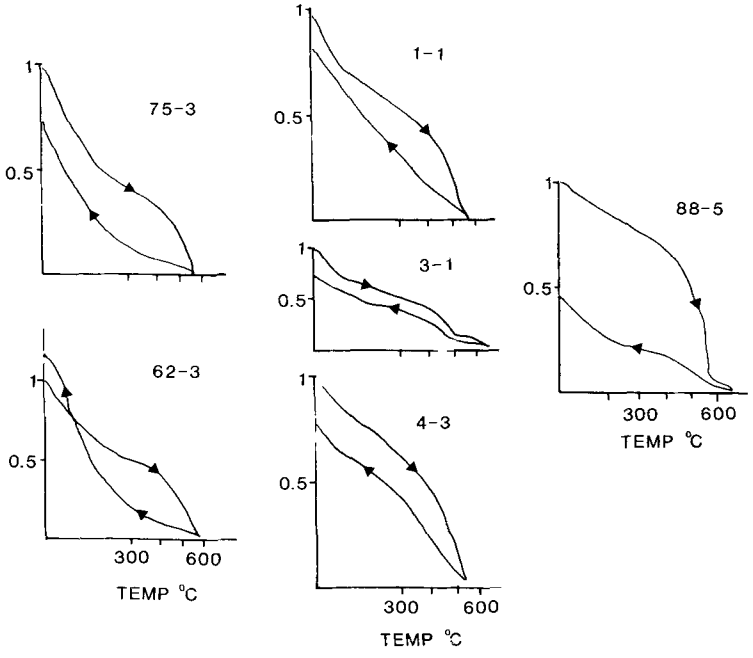


Figure 5. Thermomagnetic curves for samples exhibiting a variety of response, heating and cooling was carried out in air. All samples show a marked decrease in magnetization up to 300°–350 °C and yield Curie points in the range 550–580 °C.

magnetites observed in thin section. This result may suggest that either exsolution lamellae in the titanomagnetites or exsolved Fe oxide blades on the silicate minerals are dominating the stable remanence of the samples, neither of these possibilities has however been identified optically or on SEM images. It must be noted however that the modified Lowrie–Fuller test is not infallible and that it has been shown with prepared samples of titanomagnetites that md grains may give sd type results (Harstra 1982; Dunlop 1983).

Of 42 samples subjected to low temperature susceptibility measurements, 40 gave results indicating multidomain grain size in all ultramafite rock types no matter which characteristic direction was present in the sample. The other two samples yielded single domain results; one came from a dunite and the other from an antigorite-bearing rock and both carried an *A* component. These results therefore, reflect the abundant multidomain grain size of the magnetite seen in thin section. It has been suggested by Lienert & Wasilewski (1979) that Ni_3Fe is a common carrier of remanence in this type of ultramafites and that this mineral has similar properties to magnetite but is distinguishable from magnetite on the grounds of its higher (590 °C) Curie point. No Curie points above 580 °C were noted in this study suggesting that this phase is not present in significant amounts in these ultramafites.

The net result of all these experiments was that no detectable difference in magnetic characteristics exists between different rock types carrying the same component direction or the same rock type carrying different magnetic components. This conclusion clearly fails to isolate any reason as to why a sample should carry a particular remanence.

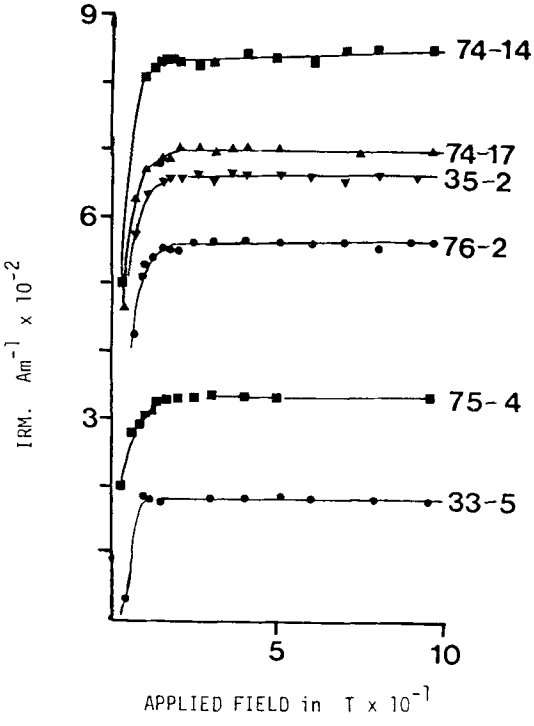


Figure 6. Typical examples of normalised IRM acquisition curves for a range of rock types and characteristic components. All show rapid saturation between 150 and 200 mT and virtually no increase above this, indicating the presence of magnetite and absence of heamatite.

4 INTERPRETATION OF RESULTS

As can be seen from Table 2 and Fig. 7a, three separate populations of directions of magnetization are recognised. These yield characteristic mean directions classed as *A*, *B*,

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Table 2. Summary of all results.

Site	Lithology	*N*	Dec	Inc	Á95	*k*	Comp	Notes
Sites from the lower nappe of Unst								
1	Antigorite	3	70.5	82.6	7.7	256.4	A	3
3/4		5	78.7	73.9	15.9	24.1	A	2, 5
23		7	82.8	54.1	7.4	67.7	A	
25		5	267.0	43.6	16.5	22.4	?	3
55		3	67.9	48.0	15.0	68.3	A	1, 2
60	Harzburgite	4	189.6	−26.3	14.0	30.7	C	4, 5
61		7	183.1	−14.9	3.9	172.8	C	
74A		9	180.0	9.6	16.5	10.7	C	1, 2, 4
76		8	188.5	−27.5	11.5	24.1	C	1, 2, 4
88		9	192.4	−0.1	14.7	9.7	C	
56	Dunite	6	87.9	50.2	23.9	8.8	A	
57		7	109.3	70.4	19.1	11.0	A	2
59		7	109.3	66.1	13.7	20.3	A	
62		7	179.0	56.5	19.7	10.3	?	
74B		10	140.0	87.7	8.0	37.7	A	
75		8	69.1	66.1	10.5	19.8	A	3
5	Pyroxenite	8	66.9	70.8	13.8	17.1	A	
30		5	69.6	78.0	16.2	23.4	A	
31		5	65.4	51.9	15.4	25.6	A	
58		7	140.4	78.0	4.8	156.7	A	
73		3	84.6	62.2	6.7	83.1	A	
73		4	23.5	10.4	21.2	9.2	B	
77		8	64.5	49.9	17.7	8.4	A	
78		7	66.2	55.1	7.3	68.4	A	
78		3	52.1	9.4	15.2	26.3	B	
85		7	118.3	62.3	17.3	13.1	A	
86		8	45.6	−31.1	11.4	17.1	B	
50	Gabbro	4	182.7	17.5	28.0	11.7	C	??
84	Sheeted dyke	12	184.1	−3.2	25.5	3.9	C	??
Sites from the upper nappe of Unst.								
32	Harzburgite	7	63.2	70.0	7.6	65.4	A	
33		7	76.7	69.5	10.2	36.0	A	
34		8	87.7	66.4	13.3	18.2	A	
35		6	188.4	50.9	10.7	39.9	?	
87	Dunite	5	142.4	−31.4	24.5	10.7	?	
Sites from the upper and lower nappes of Fetlar								
68	Antigorite	5	115.4	45.0	25.8	9.8	A	
69	Harzburgite	7	24.5	60.8	14.6	18.1	A	
70		8	127.5	80.4	15.9	10.2	A	
71		7	4.4	79.4	8.3	53.6	A	
72		7	64.3	58.8	14.0	19.5	A	
Sites from the lamprophyre dyke swarm								
82	Lamprophyre	5	193.1	41.6	7.1	93.7	?	

Notes

1 Some samples yielded *B* components, not included in site mean.
2 Some samples yielded *A'* components, not included in site mean.
3 Some samples yielded *C* components, not included in site mean.
4 Some samples yielded *A* components, not included in site mean.
5 Some samples yielded unclassified components, not included in the site mean calculation.

A' is the antiparallel direction to *A*. Sites 3 and 4 were drilled some 10 m apart and because large numbers of the collected samples gave varying directions the two sites have been combined to give one result. Site 74 was an exceptionally large site and as the differing lithologies gave differing results these have been presented separately. *N* is the number of individual samples that have given results, A95 is the radius of the 95 per cent cone of confidence about the mean direction and *k* is Fisher's precision parameter ($k = (N-1)/(N-R)$, where *R* is the length of the resultant vector).

and *C* which are:

A: easterly steep positively inclined direction, ($D = 79.9°$, $I = 68.1°$, $k = 26$, from 25 sites),

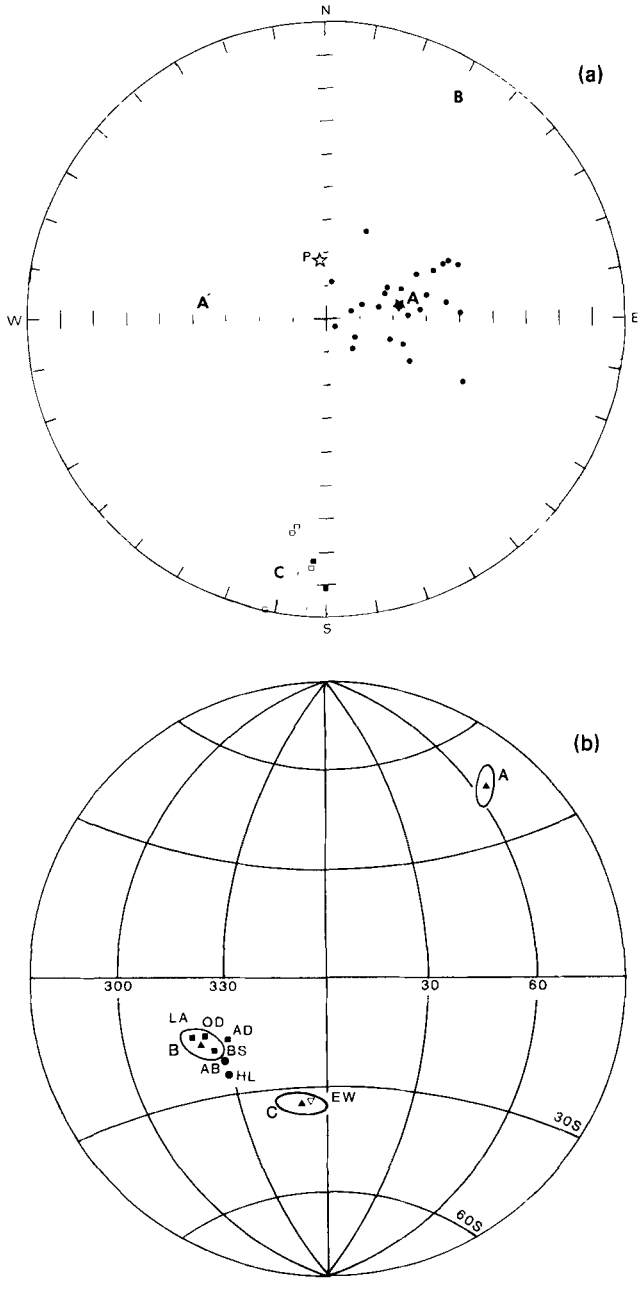


Figure 7. (a) Stereonet projection of site mean directions for *A* and *C* components, lower (upper) hemisphere directions are shown by closed (open) symbols. Also shown is the present day field direction (*P*) and the mean directions for the *A*, *B* and *C* directions and the reversed *A* component direction. (b) Lambert equal area projection of the pole positions and their 95 per cent confidence limits from this study (*A*, *B* and *C*) and those listed in Table 4. Abbreviations are: LA Loch Ailsh; OD other dykes; AD Achmelvich Dyke; BS Borrolan Syenite; HL Hoy Lavas; AB average 'B' group; EW East–West Carboniferous dyke swarm.

B: a northeast shallow negatively inclined direction, ($D = 34.0°$, $I = -12.2°$, $k = 10$, from 4 sites),
C: a southerly shallow negatively inclined direction, ($D = 186.7°$, $I = -11.9°$, $k = 23$, from 5 sites).

The *A* direction is the dominant component resolved from samples of the antigorite, dunite and clinopyroxenite facies

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of the lower nappe on Unst, from the predominantly harzburgite ultramafites of the upper nappe of Clibberswick on Unst, and in samples from both lower and upper nappes of Fetlar. The *B* direction has been identified most commonly in the pyroxenite rocks of south-central Unst (sites 73, 78 and 86) although isolated samples from other ultramafite facies have also yielded this direction. The *C* direction was found almost exclusively in the harzburgite of the lower nappe in northern Unst suggesting that some (so far unidentified) lithological and/or areal factor controlled the acquisition of this component of magnetization.

A number of samples yielded stable directions antiparallel to the main *A* direction, that is westerly and moderately to steeply inclined negatively. These samples all came from sites where the majority of samples yielded the typical *A* component directions. A number of sites (25 and 62 from the lower nappe and 35 and 87 from the upper nappe of Unst) and individual samples from a variety of sites yielded stable single component directions which cannot be reconciled with the three main directions identified. No significant difference in AF or thermal demagnetization characteristics was noted between samples carrying *A*, *B* and *C* component directions and hence all three are regarded as having similar stability.

5 THE POLE POSITIONS

Considering the undoubtably complex tectonic history of the ophiolite it is hardly surprising that it records at least three separate magnetization directions. Essentially three distinct episodes in the history of the complex are likely to have been capable of giving rise to conditions leading to the acquisition of a magnetization. These are broadly defined as, (i) pre-emplacement (i.e. in an oceanic context), (ii) obduction and thrust emplacement of the complex and (iii) post-emplacement uplift and cooling. In addition Mesozoic rifting of the North Sea in mid-Jurassic times has previously been suggested as a possible period of widespread remagnetization in this area (Storetvedt *et al.* 1978). Radiometric dating (Taylor 1985) suggests that obduction of the complex occurred in late Cambrian or early Ordovician times and that it was subjected (as was the surrounding Dalradian basement and much of Shetland) to uplift and cooling during the interval 420–430 Ma (Miller & Flinn 1966; Taylor 1985). This indicates that the actual emplacement of the ophiolite complex onto the already-metamorphosed Dalradian rocks of the area took place in Ordovician times. The youngest event which has been tentatively recognized by isotopic dating is 385 Ma, and if this was sufficiently intense to reset the K–Ar isotopic system it is quite conceivable that a partial TRM or CRM could have been imparted at this time.

The essence of this discussion is that any primary tectonothermal events capable of conditions in which a magnetization is likely to have been generated is Ordovician to Devonian in age. (A younger magnetization linked to Mesozoic rifting is unlikely both because it is not recorded by the radiometric signature and because the *A*, *B* and *C* directions do not correspond to the established Mesozoic magnetic field directions for this area). It is therefore with the Palaeozoic part of the APW path for northern Britain that the palaeopoles must be compared.

Table 3. Palaeomagnetic summary statistics.

Direction							Palaeomagnetic pole			
Comp.	*N*	*R*	*D*	*I*	*k*	Á95	Lat. S	Long. E	*dm*	*dp*
A	25	24.1	79.9	68.1	26	5.8	−47.2	64.2	8.2	9.7
A$^+$	17	15.6	271.4	−56.7	12	10.8				
B	4	3.7	35.0	−6.9	10	30.4	20.3	321.6	15.4	30.6
B$^+$	22	18.7	34.0	−12.2	6	13.4	18.0	323.4	6.9	13.6
C	5	4.8	186.7	−11.9	23	16.4	35.0	351.1	6.5	16.7

N is the number of site or sample vectors used in the calculation; *R* is vector sum; *k* is Fisher's (1953) precision estimate; Á95 is the semi-angle of the cone of confidence about the mean direction; *dp*, *dm* are the semi-axes oval of 95 per cent confidence about the mean pole; $^+$ Indicates unit weight was given to samples rather than sites.

The *A* component and palaeopole

The *A* component is the dominant magnetization in most of the ultramafite lithologies and occurs in both the upper and lower thrust nappes on both Unst and Fetlar. It must therefore post-date the imbrication of the thrust nappes, althought the slightly greater spread in declinations (4.4°–127.5°) of the sites (68–72) on Fetlar, as opposed to that seen on Unst (64.5°–140.4°), may indicate that it was acquired prior to or during the episode that folded the thrust planes themselves in this area. Although it is relatively close to the present geomagnetic field direction it is not a viscous magnetization as is clearly demonstrated by the identification of samples carrying an antiparallel direction (Table 3) and by its survival to relatively high AF fields and temperatures. The corresponding pole must consequently be regarded as an ancient one. This palaeopole does not however coincide with the generally accepted Palaeozoic or younger poles of northern Britain. Indeed the only similar pole position from rocks of comparable age to the Unst ophiolite comes from Ordovician limestones in Southern Sweden, the author however, was unable to completely verify that this magnetization was of primary Ordovician age (Claesson 1978). There are then three possibilities that might be advanced for the explanation of such a pole, derived as it is from a stable, single component remanence of dual polarity. These are:

(1) The *A* palaeopole represents a part of the APW path for lower Palaeozoic times that has not yet been recognised. If this is the case then it is most likely to be of Ordovician age as this is the least constrained time period. Inclusion of such a pole into the APW path demands a radical new path to that previously supposed, although this may yet prove to be tenable (Piper, pers. comm.).

(2a) The complex as a whole and hence also the magnetization direction have been rotated about a non-vertical axis leading to the displacement of the palaeopole from the APW path. Such a hypothesis is however untestable.

(2b) The complex is a far travelled displaced terrane, such that it has undergone translation subsequent to thrust stacking of the nappes and acquisition of the *A* component magnetization. To bring the pole position into agreement with other Ordovician data would require an enormous lateral displacement on strike slip faults relative to the forelands of the Caledonian orogen and is therefore considered highly unlikely.

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(3) A third possibility is that this magnetization does not record a geocentric axial dipolar field. Such a hypothesis has been previously advanced to explain similar steeply inclined directions of Ordovician age from the Caledonian terranes of southern Britain (Thomas & Briden 1976; Piper *et al.* 1978; Piper 1979). These authors favoured a long standing non-dipolar field, as opposed to a true intermediate field direction during a reversal, as a plausible explanation of their apparently anomalous Ordovician palaeomagnetic results. It must be pointed out however, that while there is some similarity between these results and those of this study in that all have steep inclinations, there is no overlap in the pole positions. Furthermore, these previous studies and this one of the Unst ophiolite comprise directions of both polarities; to produce directions which are closely antiparallel, a long standing non-dipolar source would need to have a complex and highly implausible nature.

At this time none of these hypotheses can be substantiated and the problem remains that while the magnetization is clearly well defined and of palaeomagnetic origin, no simple answer can be advanced to explain it's origin.

The *B* and *C* components and their corresponding poles

Both of these directions and their corresponding palaeopoles are comparable with results from mainland Scotland. The data corresponding most closely with the *B* component are shown in Table 4 and Fig. 7(b) and are divided into two distinct groups; Group 1 are results from the NW alkali syenite intrusions of Loch Borrolan, this complex is dated at 430 Ma (U-Pb) and the magnetizations are regarded as primary (Turnell & Briden 1983); Group 2 results are from the Orcadian Basin. The age of these Orcadian Basin results is rather controversial with Storetvedt and his co-workers supporting a middle to upper Devonian primary/early diagenetic age (e.g. Storetvedt & Meland 1985) while Tarling (1985) and others prefer a middle Carboniferous overprint age for this direction arguing in effect that almost all results from the Orcadian basin are dominated by chemical overprint remanences. Both these interpretations however, require that the inferred APW path recrossed the earlier position indicated by the Group 1 results. The *B* pole thus falls into an ambiguous area. The choice of a Siluro–Devonian age is consistent with known tectonism in the Shetland area but an upper Devonian or Carboniferous age cannot be excluded.

Table 4. A summary of selected pole positions comparable to those identified in this study

Locality	*D*	*I*	Á95	Lat. S	Long. E	Ref.
Group 1: Poles from the NW Scotland alkali igneous suite						
Loch Ailsh	33.1	−21.6	7.6	15.6	321.3	1
Achmelvich Dyke	23.6	−24.2	3.0	16.5	330.6	1
Other dykes	29.9	−23.3	9.7	15.5	324.5	1
Boralan Syn.	26.9	−19.4	7.5	18.4	326.9	1
Group 2: Poles from the Orcadian Basin						
Average of *B* directions	203	7	10	26	330	2
Hoy lavas	208	9	5.5	22.7	329.5	3
Poles comparable to the Unst *C* pole						
EW dykes Argyllshire	180.1	−3.8	4.9	35.0	354.6	4

Refs 1 Turnell & Briden (1983)
2 Tarling (1986)
3 Storetvedt & Meland (1985)
4 Esang & Piper (1984)
Notation as in Table 3.

The *C* pole in comparison to the *A* and *B* poles would appear to be more simply interpreted in that it overlaps the well defined result from the Carboniferous EW dyke swarm of Argyll *ca.* 320 Ma (Esang & Piper 1984) which itself is consistent with the APW path for continental Europe at this time. Accepting this age (close to the lower–upper Carboniferous boundary) for this pole would tend to mitigate against a Carboniferous age for the Orcadian Basin magnetizations argued for by Tarling and would imply that the *B* pole is more likely to be older than the *C* pole and be either of middle to upper Devonian age or Siluro–Devonian in age.

6 CONCLUSIONS

Three distinct magnetizations have been isolated from the ultramafite units of the Shetland ophiolite complex. All three have similar magnetic properties and showed no preferential stability with respect to each other. The broad range of temperature blocking spectra and low to medium median destruction fields would tend to suggest that all three magnetizations have similar origins. The most likely mechanism for producing these magnetizations in these rocks is thermo-chemical magnetization caused by serpentinisation, or alteration of the magnetite originally produced by serpentinisation. This would explain why only the serpentinised ultramafites in this complex carry a significant remanence. The dominant *A* component is most likely to be the oldest remanence and to have an Ordovician age because it does not correlate with the post-lower Silurian field in this region, but this cannot be confirmed as yet. The *B* component direction has been assigned a tentative Siluro–Devonian age and the *C* component a mid-Carboniferous age. Two of the four sites with stable remanence directions (sites 35 and 62) that were excluded from the overall mean directional analysis have directions close to components attributed to a primary upper–middle ORS (upper Devonian–lower Carboniferous) origin by Robinson (1985). Clearly palaeomagnetism has detected a number of subtle events of differing age which appear to be of thermo-chemical origin as opposed to primary TRM. This suggests that the processes responsible for serpentinisation of the complex continued to occur long after tectonic emplacement, or that subtle low-temperature alteration events affected the magnetite grains initially formed by serpentinisation. Such alteration events are likely to have been due to either tectonic or climatic processes, for example Devonian strike-slip motions or permo-Carboniferous weathering regimes.

ACKNOWLEDGMENTS

The author thanks Dr J. D. A. Piper both for his assistance during fieldwork and for reviewing the manuscript at an

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earlier stage in the preparation. Much of this work was carried out in the Sub-Department of Geophysics, University of Liverpool and I thank all those who gave advice and assistance. I am particularly grateful to Mr and Mrs W. Jamieson for their help and hospitality in Shetland, This work was supported by a NERC studentship.

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